

Examiner reports

Q1.

This question was intended to discover whether the student had progressed beyond the type of investigation which might have been carried out at GCSE. There were a minority of answers where the student identified the central features, showed what happened with endings of 1, 3, 7 and 9, and then completed the proof efficiently. Some of these were very elegant and brief, but this was not necessary to earn the R1 mark for mathematical rigour.

Many looked at 7^4 , 11^4 and 13^4 , and often many more examples, failing to realise that this approach in itself would never prove anything.

Q2.

Part (a) was well attempted and many students scored full marks. The question left students with the choice of how to approach the proof and inefficient but correct methods were often seen. Students who scored one mark often failed to complete the proof. Students should be in the habit of finishing any proof or “show that” question with a concluding statement which completes their argument.

While many students scored full marks on part (b) the most efficient method was rarely seen. Students who solved $p(x) = 0$ on their calculators and made use of the factor theorem to write down the correct factorisation could achieve full marks in one line.

Part (c) discriminated well between students of varying abilities. Many students could start the question by manipulating the given equation to form a cubic equation. Some students who went on to see the connection with the earlier part of the question made errors with their use of inequalities or failed to complete their argument with enough rigor.

Some students seemed to focus on the phrase “no real solutions” and were determined to try and use the discriminant of an incorrect quadratic.

Q3.

This was an easy question, which was answered correctly by a very high proportion of students. The most frequent incorrect selection was $x^2 = 4 \Leftrightarrow x = 2$.

Q4.

Most students realised that proof by exhaustion was required and the majority achieved the first mark for starting to check for factors. The second mark could only be achieved through a rigorous proof and a variety of approaches ranging from very efficient to very inefficient were seen. Some students checked every integer between 1 and 23 with the consequence that missing out one number lost a mark. Others explained they only needed to check prime numbers and a few realised they only needed to check the two prime numbers less than and, provided this was explained, full credit was given.

Q5.

This question generally produced very strong or very weak responses. Several students had memorised the proof very clearly. A number of students were able to set up the proof but got lost trying to show that a and b were even, whilst weaker students were unable to make any progress.



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